

A Robust Wavelet-Based Digital Watermarking Using Level-Adaptive Thresholding

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Abstract

In this paper, a robust digital watermarking algorithm using wavelet transform is proposed. In order to equally embed the watermark to the whole image, the coefficients of all subbands including LL subband are utilized. To select perceptually significant coefficients for each subband, a level-adaptive thresholding scheme is used. The watermark is embedded to the selected coefficients, using different scale factors depending on the level of decomposition. In the detection process, the vector projection method is used. The performance of the proposed algorithm compared with other transform-domain watermarking methods has been analyzed, using various test images. The experimental results have shown that the proposed watermark is invisible to human eyes and very robust to various attacks, such as image compression, image filtering, geometric transformations and noises.

1. Introduction

Recently various kinds of digital multimedia data have been widely distributed through internet. As the commercial price of multimedia data is increased and the resulting multimedia data can be easily duplicated via network, the importance of copyright protection for multimedia data becomes more crucial. To solve this problem, the digital watermarking techniques have been investigated[1].

Digital watermarking has been studied in the spatial domain and frequency domain. The spatial-domain watermarking scheme is generally fast and simple, but it doesn't guarantee that the watermarking would be robust against noises and JPEG compression. As a frequency-domain method, Cox et al. proposed to embed the watermark into only top (largest in absolute magnitude) thousand coefficients excluding the DC coefficient, after transforming the original image using DCT[2]. Xia et al. proposed a frequency-domain method of embedding the

watermark at all the subbands except LL subband, using DWT[3]. The frequency-domain watermarking methods are relatively robust to noise, common image processing and compression, compared to the spatial-domain methods.

In this paper, we propose a new digital watermarking method that effectively embeds the watermark at perceptually significant wavelet coefficients. The proposed method generates the watermark using a gaussian distributed random vector. In the insertion process, the coefficients of all subbands including LL subband are utilized in order to equally embed the watermark to the whole image. To select perceptually significant coefficients for each subband, we use level-adaptive thresholding. Then the watermark is embedded to the selected coefficients using different scale factors depending on the level of decomposition. In the detection process, the vector projection is used to calculate the similarity between the original watermark and the extracted watermark.

2. The Proposed Watermarking Algorithm

The overall scheme of the proposed watermarking algorithm is shown in Figure 1 and 2.

First, the watermark is generated using Box-Muller transform. The perceptually significant coefficients are selected among wavelet coefficients using multi-level thresholding. The watermark is embedded to the selected coefficients, using different scale factors according to the level of decomposition. In the detection process, the vector projection method is used to calculate the similarity between two vectors.

2.1. Generation of Watermark

Since wavelet coefficients can be generally characterized by the gaussian distribution[4, 5], the proposed algorithm uses a gaussian distributed random vector as the watermark for invisibility and robustness. By

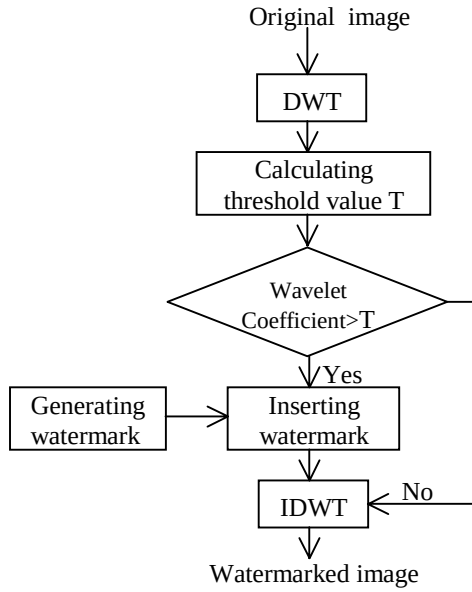


Figure 1 : Watermark insertion process

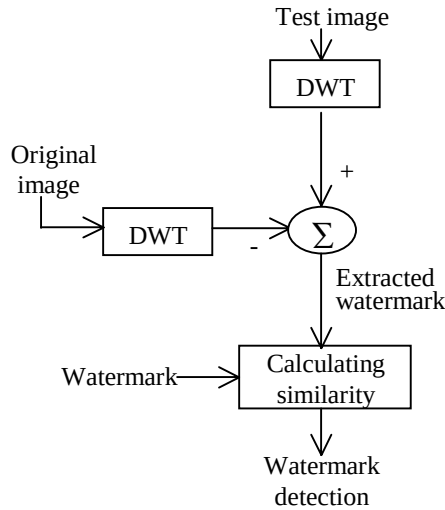


Figure 2 : Watermark detection process

using Box-Muller transform, a gaussian distributed random vector X_i is generated from a uniform distributed vector E_i , as defined in equation (1).

$$X_i = \sqrt{-2 \ln E_i} \cos(2\pi E_i + N) \quad (1)$$

where N is the number of watermarks to be generated.

2.2. Watermark Insertion

Watermark insertion process is composed of DWT of the original image, calculation of multi-level thresholds for selecting perceptually significant coefficients, watermark

insertion, and IDWT of the coefficients with watermarks. Figure 1 illustrates the overall process of watermark insertion.

To find the perceptually significant wavelet coefficients for each subband, the threshold value is calculated according to the decomposition level. For example, in the 3-level decomposition, the largest coefficients C_1 for 1-level subbands(LH₁, HL₁, HH₁) is selected and the threshold T_1 is calculated by equation (2). T_2 and T_3 for the subsequent levels are respectively calculated using the same procedure.

$$T_i = 2^{\lfloor \log_2 C_i \rfloor - 1} \quad (2)$$

where i is the decomposition level and $\lfloor X \rfloor$ represents the largest integer which is no greater than X . The watermark is embedded only to the selected coefficients.

To the selected coefficients, the watermark is embedded using equation (3). It makes the watermark adaptively inserted according to the coefficient value. The larger the coefficient value, the larger the inserted watermark.

$$V'_i = V_i + \alpha V_i X_i \quad (3)$$

where V_i is the selected wavelet coefficient and X_i is a gaussian distributed random vector.

The proposed method uses an appropriate value of scale factor α for each subband. Since the values of wavelet coefficients for LL subband are generally large, a small scale factor of 0.04 is used to avoid any severe quality degradation of watermarked image. For other subbands, the scale factor is properly tuned according to the decomposition level. Since the mean of wavelet coefficients is reduced by a half as one level goes up, scale factor α is increased twice. This adaptive scale factor improves the performance of robustness and invisibility. In this paper, scale factors of 0.1, 0.2, and 0.4 are used for level 3, 2, and 1, respectively.

2.3. Watermark Detection

The detection process is inverse procedure of the watermark insertion process. It is composed of DWT of original image and watermarked image, watermark extraction, and calculation of the similarity. The proposed algorithm extracts the watermark using the original image. Figure 2 illustrates the overall process of watermark detection.

To extract the watermark from the watermarked image, the original and the watermarked image are transformed

using DWT. Since the watermark is the difference between the original image and the watermarked image, we subtract wavelet transformed subbands of the original image from the watermarked subbands.

The similarity between the original watermark and the extracted watermark is calculated to detect the existence of watermark. The similarity should be increased or decreased according to the amount of the inserted watermark, as well as to the degree of the attack.

To calculate the similarity, the proposed method utilizes the vector projection as defined in equation (4).

$$sim_1(X, X^*) = \frac{X \cdot X^*}{\sqrt{X^* \cdot X^*}} \quad (4)$$

where X is the original watermark and X^* is the extracted watermark. The value calculated from equation (4) may change, depending on the length of the vector. Therefore, we also use another measure of similarity, which is the normalization of equation (4). Equation (5) means the percentage of the remained watermark against the attack.

$$sim_2(X, X^*) = \frac{X \cdot X^*}{\sqrt{X^* \cdot X^*}} / \frac{X \cdot X}{\sqrt{X \cdot X}} \times 100 \quad (5)$$

where $\frac{X \cdot X}{\sqrt{X \cdot X}}$ is the self-similarity.

If the similarity value is greater than a threshold value in equation (6), we detect the existence of watermark

$$S = \frac{\alpha}{2M} \sum |V_i| \quad (6)$$

where M is the number of coefficients where the watermark is inserted and α is the scale factor used in the watermark insertion.

3. Experimental Results

The proposed method uses biorthogonal discrete wavelet transform to decompose the original image into 3 levels. The length of watermark is about 1000. The test images include “Lena”, “Cameraman”, “Baboons”, and “Airplane” image of size 256×256 . The performance measure is the invisibility and the robustness.

By using the proposed method, the watermark is almost invisible to the human eyes, as shown in Table 1.

Table 1 : PSNR of watermarked images

Image	Lena	Cameraman	Baboons	Airplane
	52.46	47.74	49.88	53.37

Figure 3 is the original “Lena” image and Figure 4 is the watermarked image. Table 1 and Figure 4 show that the image distortion after inserting the watermark using the proposed method is negligible. Figure 5 shows the response of the watermark detector to 1000 randomly generated watermarks.



Figure 3 : Original Lena image



Figure 4 : Watermarked Lena image

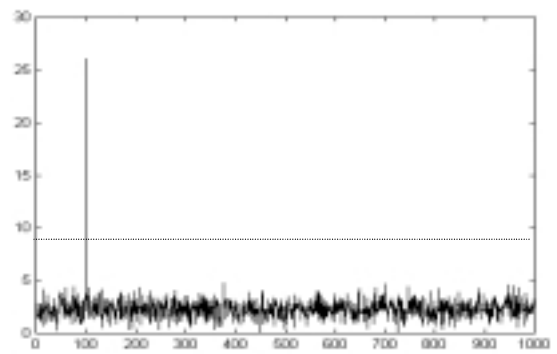


Figure 5 : Watermark detector response to 1000 randomly generated watermarks (No attack)

To verify the robustness of the proposed method, we calculate the similarity using equation (4) after attacking the watermarked image by 6 different attacks, as shown in

Table 2. Figures 6-9 show the robustness of the proposed method for all the test images and various kinds of attacks.

Table 2 : Robustness of the proposed algorithm

Image	Lena	Cameraman	Baboons	Airplane
Gaussian noise	20.35	22.54	18.45	19.52
JPEG	17.68	17.02	17.32	18.44
Wavelet compression	21.87	20.27	19.76	21.14
Averaging	19.04	20.34	18.98	20.74
Center cropping	14.43	15.05	14.33	15.67
Salt&pepper noise	13.87	13.21	13.42	12.21

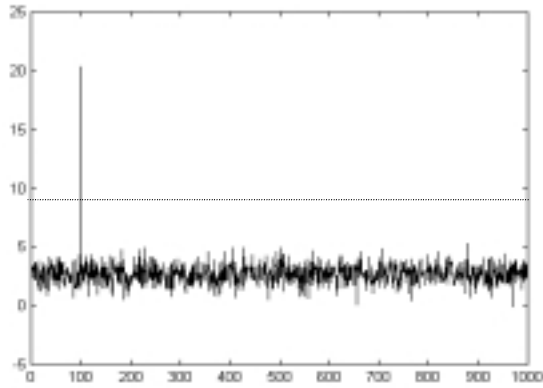


Figure 6 : Watermark detector response (attacked by gaussian noise)

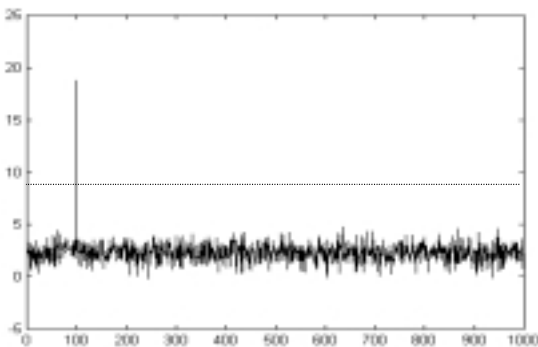


Figure 7 : Watermark detector response (attacked by JPEG compression)

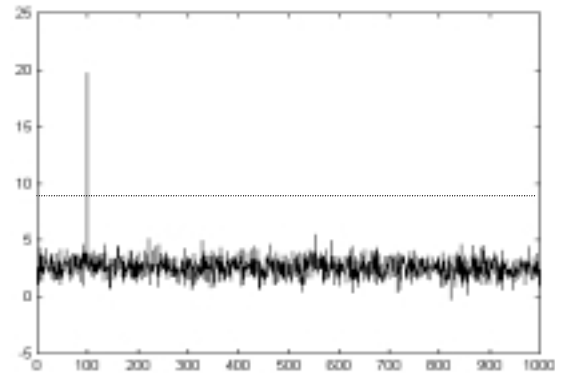


Figure 8 : Watermark detector response (attacked by averaging)

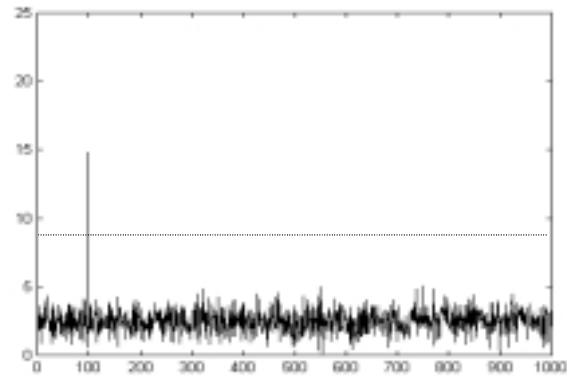


Figure 9 : Watermark detector response (attacked by center cropping)

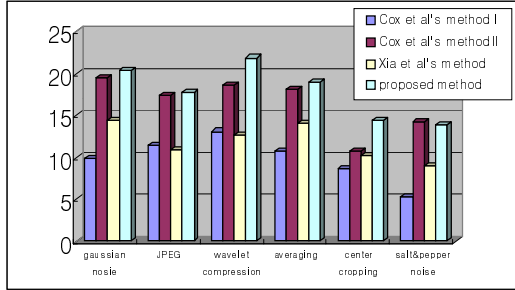
We compare the performance of the proposed method with other existing frequency-domain techniques, such as (1) Cox et al.'s method based on DCT, (2) simple extension of Cox et al.'s method using DWT, and (3) Xia et al.'s method. Table 3 shows PSNR after inserting watermark by these methods. It shows that the proposed method outperforms the others in terms of invisibility.

Table 3 : Performance comparison(invisibility)

	Cox et al.'s Method I	Cox et al.'s Method II	Xia et al.'s method	Proposed method
PSNR	51.36	42.43	50.12	52.46

Figure 10 summarizes the performance comparison using the similarity measure defined by equation (4). It shows that the proposed method is robust to all the attacks and outperforms the other methods. Cox et al.'s method II shows the similar performance to the proposed method but the image quality is degraded by 10 dB, which is noticeable to the human eyes.

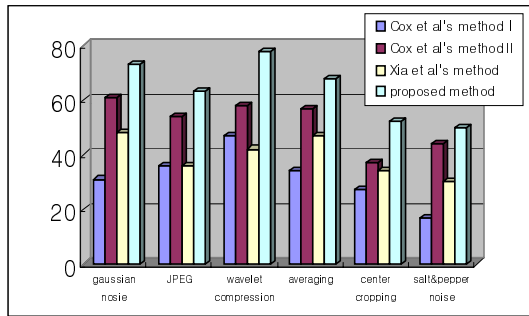
Figure 10 : Performance



comparison(robustness)

The performance comparison using the similarity measure of equation of (5) shows the similar results, as shown in Figure 11.

Figure 11 : Performance comparison



using equation (5)

3. Conclusion

In this paper, a new digital watermarking algorithm using wavelet transform is proposed. The proposed watermarking method improves the performance in terms of invisibility and robustness, by using level-adaptive thresholding to select perceptually significant coefficients and by using different scale factors according to the level of decomposition.

The proposed method has been verified to be superior to other existing transform-domain watermarking schemes, especially for highly compressed images. Future research includes the development of a robust watermarking algorithm without using the original image.

References

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